AeroDef Manufacturing

COLLABORATIVE INTELLIGENCE

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AUTOMATION-ASSISTED WORK CELLS
FOR COMPOSITE COMPONENT MANUFACTURE

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May 1, 2019 @ 11:00AM

Agenda Topics

Introduction – Impetus for Development…

Elemental Technologies

Example Work Cells

Conclusions

Question & Answer

Introduction

- Large-Scale Manufacturing Automation employed worldwide
- ‘Lights-Out’ automation employed in many industries – where ROI justifies it
- Aerospace manufacturing adoption of automation has been steadily increasing for many processes
  - Resin and fiber manufacture, impregnation, and conversion
  - Ply cutting
  - Pick-and-place ply and tape lamination
  - CNC automated fiber and tape placement
  - CNC component trimming and machining
  - and many others…
- **Accudyne Systems and Aligned Vision have been change-agents in this trend…**

Introduction

However…

Manual Layup & Inspection still widespread despite automation advancements…

- sound business case may not exist to support investment
- material systems, components, and/or processes may not lend themselves to an automated solution

The solution:

Automation-assisted work cells addressing these challenges with cost-effective solutions that judiciously integrate automation technologies.

Introduction

Automation-Assisted Work Cell Concept:

- Apply automation technology where it makes sense:
  - tasks that require more precision, repeatability, and/or heavy lifting
- Let the operators do the tasks they can do well:
  - tasks that require dexterity, hand-eye coordination, human-learning (‘the knack’)

“Human-Machine Collaboration”

Agenda Topics

Introduction

Elemental Technologies – Recipe for Success…

Example Work Cells

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Elemental Technologies

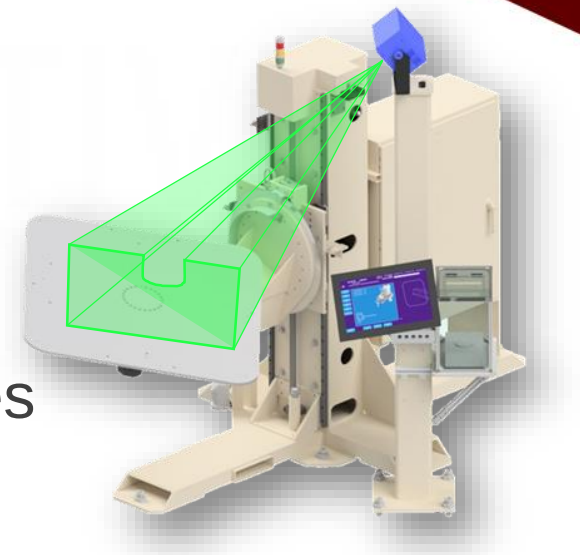
- Tool Positioning
- Integrated Laser-Projected Ply Templates
- Ply Backing Accountability
- Work Order and Recipe Management
- Manufacturing Information System Integration

Tool Positioning

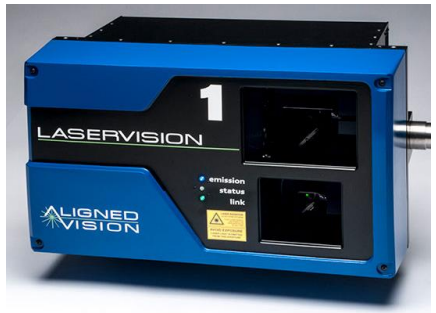
- Primary physical task interface between operator & workpiece
- Work Cells designed with ergonomics in mind for each operator
  - Properly positioned tooling reduces fatigue, muscle strain, and possible injury
- Servo-driven axes for precise control and repeatability
- Coordinate System Basis for Work Cell

Laser-Projected Ply Templates

- Throw away those assembly aid templates !
- Improve accuracy, repeatability and efficiency



- ***The Past:*** stationary tooling and Mylar templates
- ***Now:*** repositionable tooling and laser projection on same coordinate system
  - Aligned Vision integration partners have access to an ‘SDK’ to facilitate this



- ***The Near Future:*** Automatic inspection and tool/projector alignment
  - Enabling one-touch feature verification and accelerating tool re-positioning

Ply Backing Accountability

- Necessary to prevent inclusion of Foreign Object Debris
- We are all too familiar with the criticality of this feature
- A simple letterbox-style accountability system addresses this concern
- Recipe-based instructions guide the operator and HMI controls recipe advancement

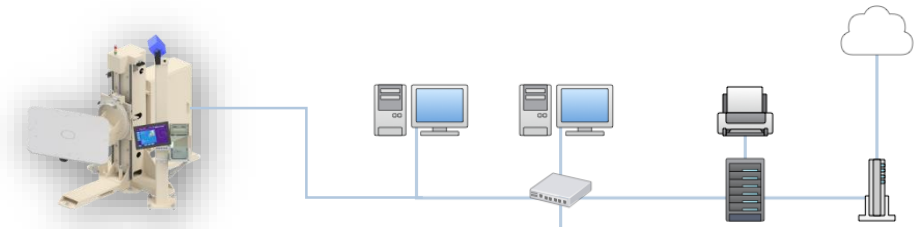
Work Order / Recipe Management

- Ties all of the integrated technologies together
- Tracks operator(s) progress throughout the component build
- Provides step-by-step standardized work instructions and prompts
- Recipe files can also contain or reference additional data
  - step number / layer / ply
  - component and/or Part Number identification
  - operation type and duration
  - component geometry references
  - laser projection geometry and/or CNC G-Code references;

- HMI incorporates Multi-User Login permissions control access to features
  - e.g. operator, engineer, maintenance, admin, etc.

MIS Integration

- Manufacturing Network-Connected Equipment
  - Version control and security of recipe, component/laser projection geometry, G-Code
- Efficient tracking of scheduled, in-progress and completed tasks & W/O
  - SQL database architecture
- Firewalled remote-access permits HMI and PLC reprogramming
  - Feature additions, remote-diagnosis, and software updates without costly delays



Agenda Topics

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Elemental Technologies

Example Work Cells – Tailored to Part & Process…

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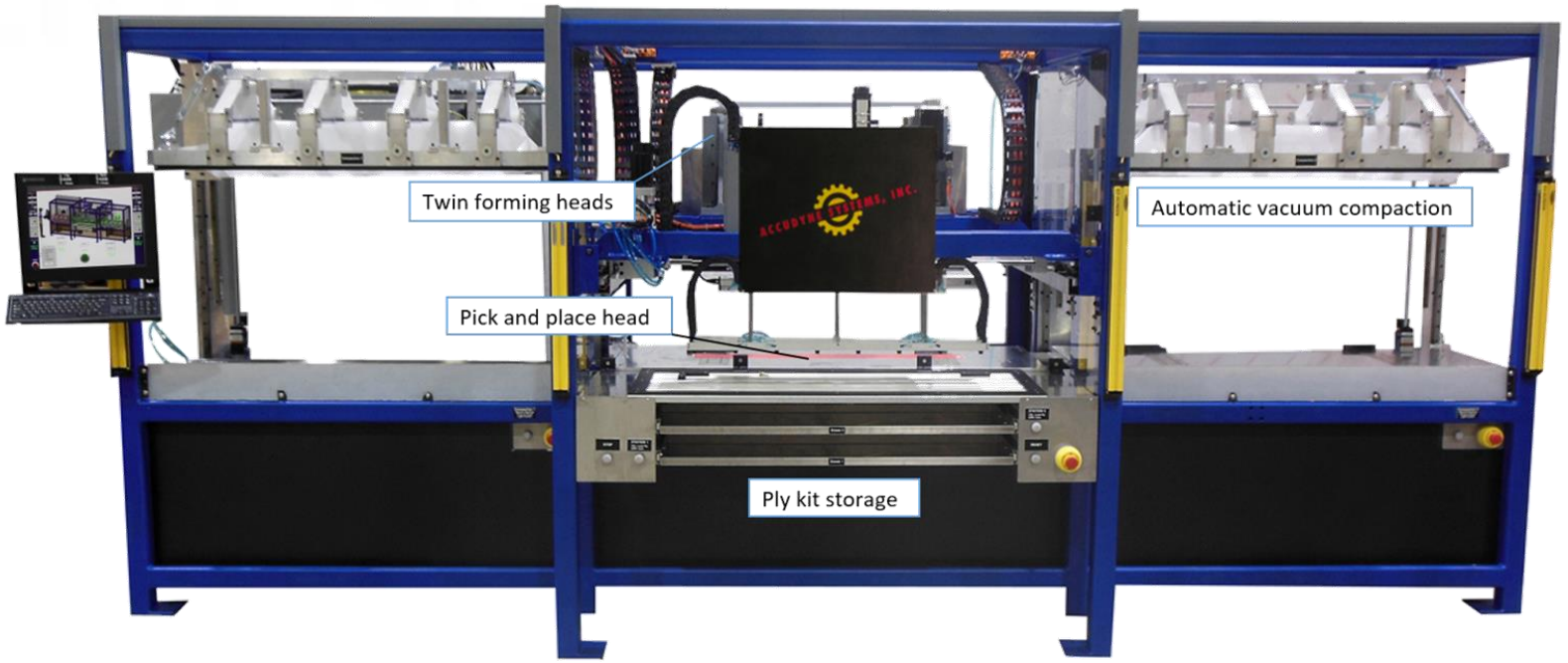
Pick/Place/Form/Compact Work Cell

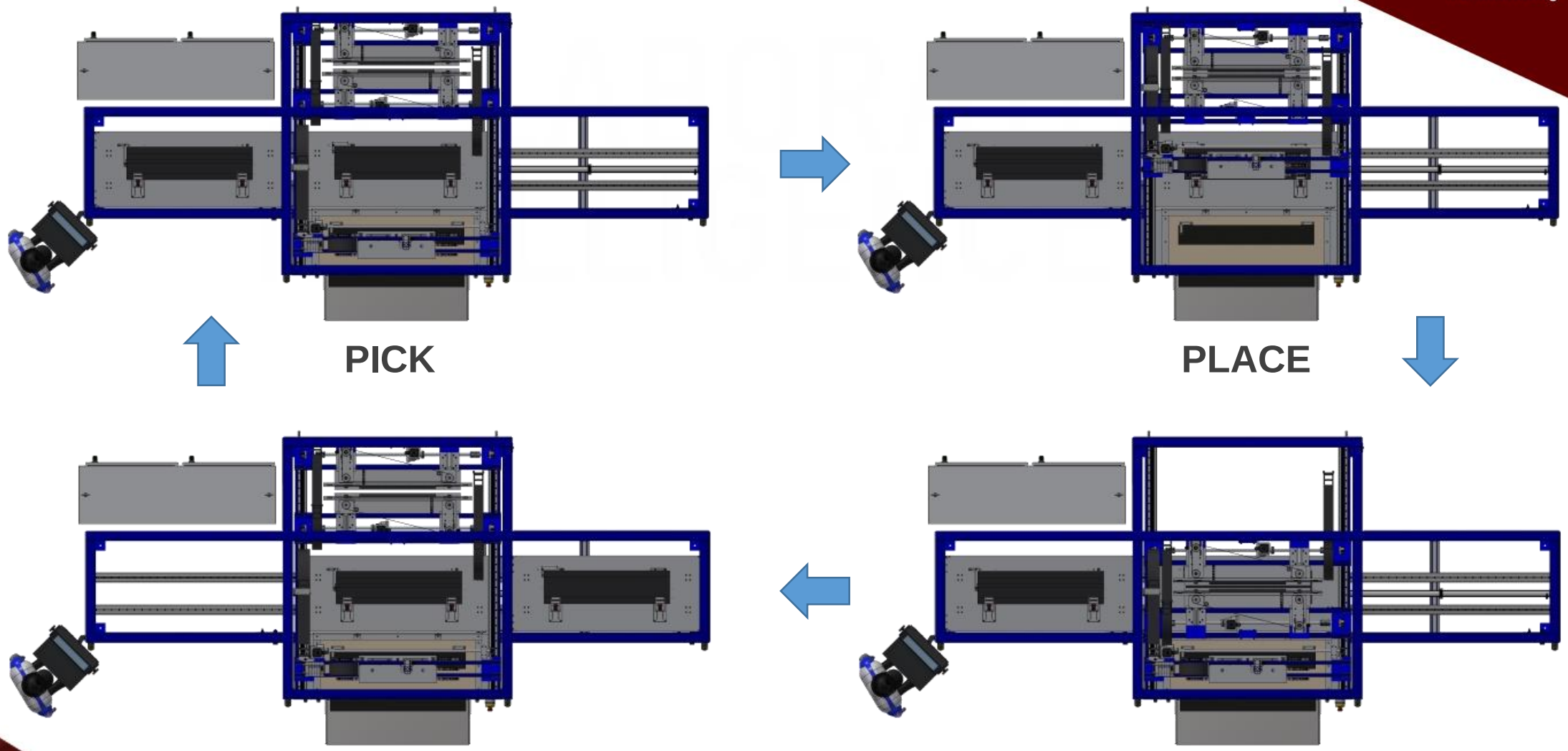
- Facilitates lay-up of two parts simultaneously
- Operator and machine work in tandem
- ‘C,’ ‘T,’ ‘Z,’ ‘L,’ and other shapes

Key Elements:

- Tool Positioning
- Pick-and-Place
- Work Order and Recipe Mgt
- MIS Integration



Pick/Place/Form/Compact Work Cell



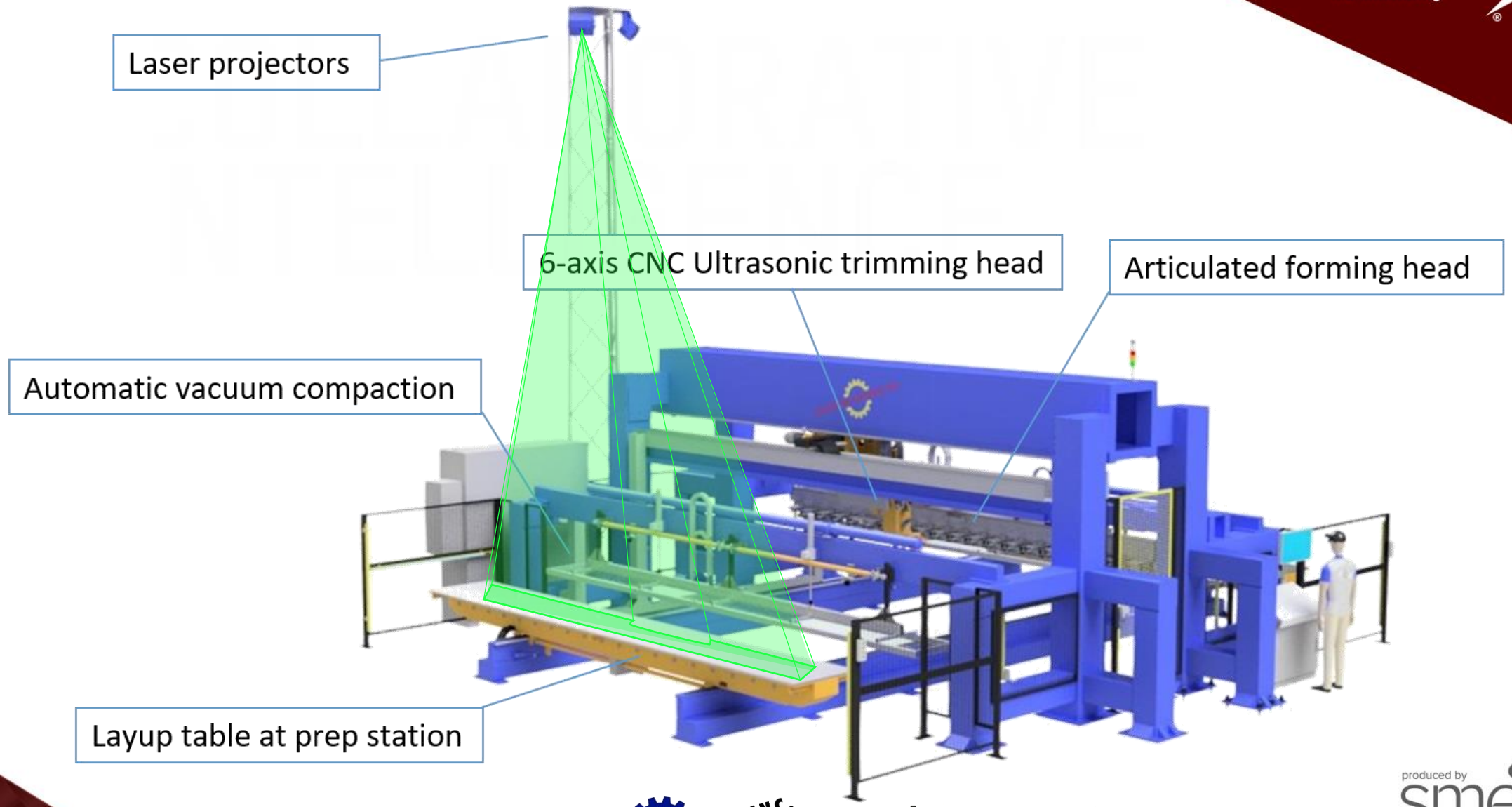
Stringer Former and Trim Cell

- Lay-up two parts simultaneously
- Tandem operator & machine tasks
- Geometry-adaptive forming mechanism
- 6-Axis CNC Ultrasonic EOP Trim

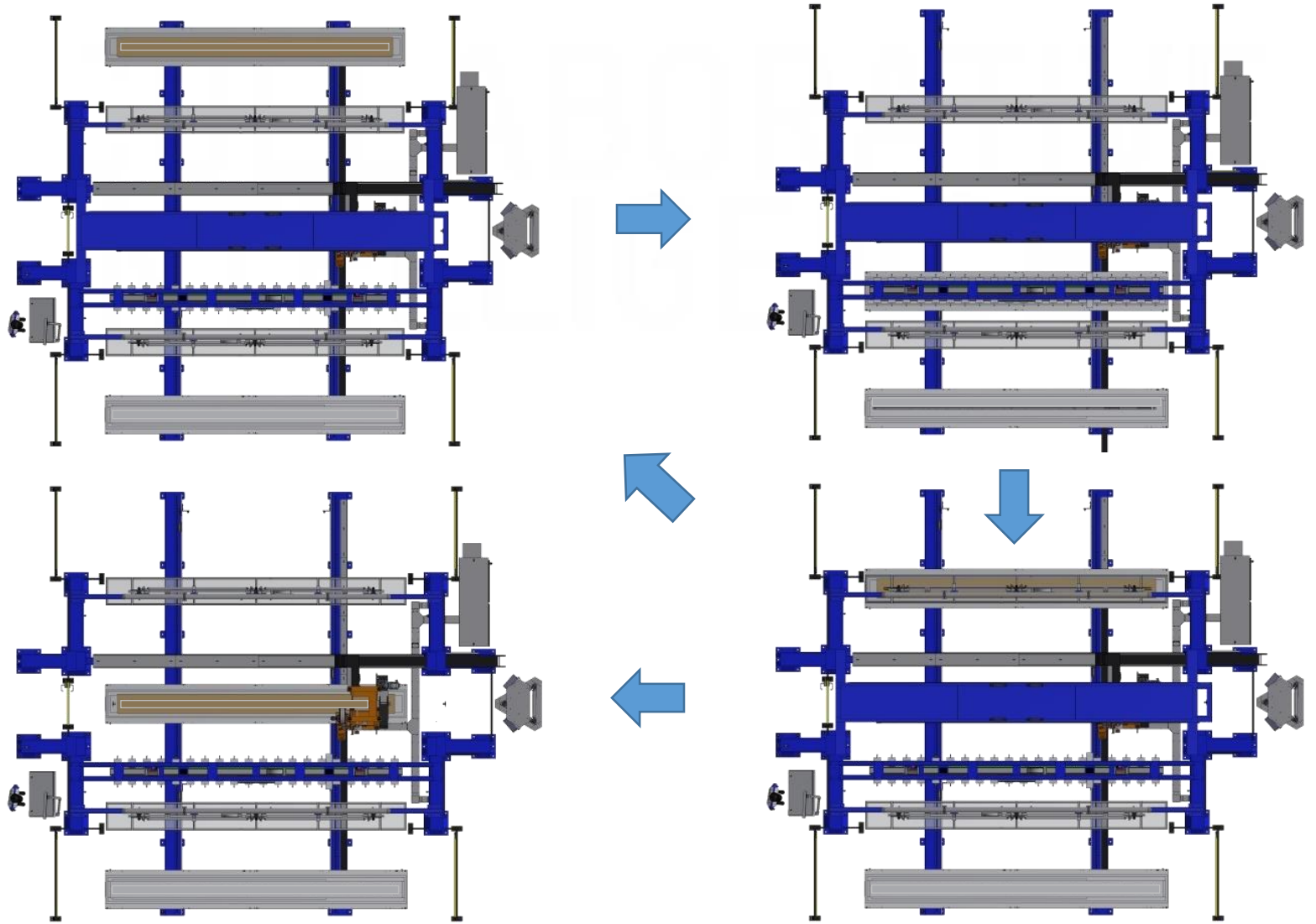
Key Elements:

- Tool Positioning
- Integrated Laser Projection (x2)
- Work Order and Recipe Mgt
- MIS Integration

Stringer Former and Trim Cell



Stringer Former and Trim Cell

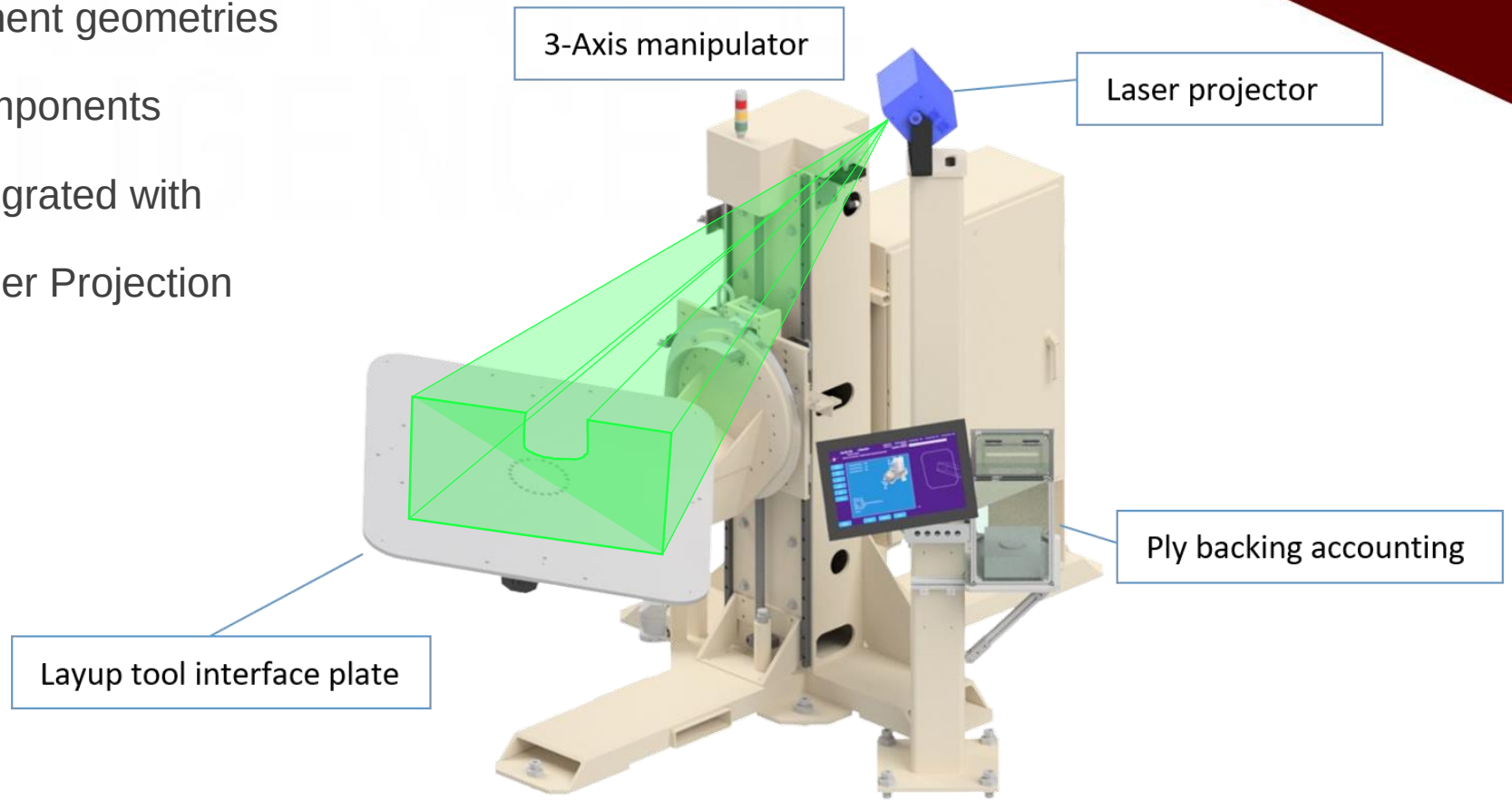


Multi-Axis Work Cell

- Single or multiple component layup
- Accommodates multiple component geometries
- Well suited to high ply-count components
- 3-Axis servo tool positioning integrated with
- Common coordinate system Laser Projection

Key Elements:

- Tool Positioning
- Integrated Laser-Projection
- Ply Backing Accountability
- Work Order and Recipe Mgt
- MIS Integration



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Conclusions – Factory Automation Innovation Continues...

Question & Answer

Conclusions

- Aerospace industry likely to continue moving toward full automation
- A ‘middle-ground’ will remain where Automation Assisted Work Cells will excel
- Automation assisted work cells have demonstrated:
  - improved process accuracy, decreased cycle time, and reduced rework and scrap rates
- Work Cells are designed to be extensible for feature additions in the future
  - i.e. integration of LASERVISION and TARGETGUIDE on work cells already equipped with LASERGUIDE
- The Automation Assisted Work Cell concept is adaptable to specific parts/processes

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Thank You for Your Attention !

What Questions do you have ?

References / Sources

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